Pythagorean Theorem

Question Bank - Additional 6.03 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 3, find the hypotenuse. Give it exactly, and to the nearest tenth when it is not a whole number.

1. Legs of 9 ft and 12 ft

15 ft

2. Legs of 3 inches and 10 inches

√109 in ≈ 10.4 in

9 + 100 = 109.

3. Legs of 6 m and 13 m

√205 m ≈ 14.3 m

36 + 169 = 205.

For questions 4 through 6, find the missing leg. Give it exactly, and to the nearest tenth when it is not a whole number.

4. Hypotenuse 10 cm, one leg 8 cm

6 cm

100 − 64 = 36.

5. Hypotenuse 7 inches, one leg 4 inches

√33 in ≈ 5.7 in

49 − 16 = 33.

6. Hypotenuse 37 m, one leg 35 m

12 m

1,369 − 1,225 = 144 = 12^2.

7. A rectangle is 6 cm wide and 8 cm long. How long is its diagonal?

10 cm

The diagonal is the hypotenuse of a 6-by-8 right triangle.

Choose one.

8. A right triangle has a hypotenuse of 26 ft and one leg of 24 ft. How long is the other leg?

- ○ A. 100 ft
- ● **B. 10 ft**
- ○ C. 2 ft
- ○ D. 35.4 ft

B

676 − 576 = 100 = 10^2.